

END TERM EXAMINATION**THIRD SEMESTER [B.TECH.] FEBRUARY-2023****Paper Code: ICT-209****Subject: Object Oriented Programming
using C++****Time: 3 Hours****Maximum Marks: 75****Note: Attempt five questions in all including Q.No1 which is compulsory. All questions carry equal marks.**Q1. Attempt **any five** of the following:-

- What is class? Differentiate between Abstract class and interface class?
- What is an object? What are the Characteristics of an object?
- Why friend functions are not advisable to use in good programming? Justify.
- What is constructor? What is the purpose of default constructor? Does constructor return any value? Is constructor inherited?
- What will be the output of the following code?

```
#include<iostream>
#include<iterator>
#include<vector>
using namespace std;
int main()
{
    vector<int> ar = { 1, 2, 3, 4, 5 };
    vector<int>::iterator ptr = ar.begin();
    ptr = advance(ptr, 2);
    cout << *ptr << endl;
    return 0;
}
```

- Justify the following statement using suitable example.
"A derived class's destructor cannot invoke its base class's destructor"

- Why Do You Need Object-Oriented Programming? Discuss its characteristics and their usage in detail.
- What is operator overloading? Explain with example the usefulness of this feature. Enlist the operators that cannot be overloaded.
- What is the difference between dynamic memory allocation and static memory allocation?
- What do you mean by 'this' pointer? How to initialize the pointer? Explain using example.
- Define state, data member, attribute, property.
- What is super, base, parent, inherited class?
- How do you make a class member friend of another class?

- ~~Q5.~~ (a) Differentiate between static binding and dynamic binding.
(b) How to resolve conflicts in naming between multiple parent classes if they are called from a child?
- ~~Q6.~~ (a) Differentiate between call by value and call by reference. Write a program to swap two integer values by using call by reference.
(b) Differentiate between compile time polymorphism and runtime polymorphism. Give suitable example.
- ~~Q7.~~ (a) Define Constructor. Differentiate between copy constructor and default constructor. Write a program to illustrate the concept of parameterized constructor.
(b). Write a program in C++ to add two complex numbers using '+' operator overloading.
- ~~Q8.~~ (a) What is exception handling? How the exceptions are handled in C++? Illustrate with an example.
(b) Differentiate between text and binary file? How they are handled differently? Give example of five functions to read and write into both types of files.